Digital Image Processing

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Submission deadline : 1405/3/16 [Image Enhancement, Image Restoration] HomeWork 4

- Students are permitted a maximum of **8 day delays** throughout the course and you are allowed to use up to 5 days of your allowed delay per homework.
- Once the allowable delay period has expired, a **20% deduction** from the total assignment score will be applied **for each additional day of delay**.
- If external resources, such as **AI tools, websites or other references**, are utilized, proper **citation and attribution** are mandatory.
- Any instance of **plagiarism or cheating** will result in a score of **0** for all parties involved. A second violation will be considered **academic dishonesty** for the entire course, with further disciplinary actions as per institutional policies.
- Please upload all your answers in a file named **HW#_[SID]_[FullName].zip** to the **CW** system.

Theoretical (60 Points)

1. What are the key differences between image enhancement and image restoration? (5 Pts)

2. A camera captures a moving object with uniform velocity in the x-direction. Exposure time is T, velocity is α (units: pixels/second), so the total displacement during exposure is $\alpha_0=\alpha T$. (10 Pts)
   a. Derive the point spread function (PSF) h(x,y) of this motion blur.
   b. Write the corresponding frequency response H(ξ1,ξ2).

3. Explain the design of an FIR Wiener filter for image restoration. (10 Pts)
   Include:
   a. The convolution form of the estimate.
   b. The orthogonality condition leading to a set of linear equations.
   c. The role of the noise-to-signal ratio σ_η^2/σ^2 in determining filter size and behavior.
   d. What happens in the two extreme cases: (i) SNR $\to\infty$, (ii) SNR $\to 0$.

4. Spatial Operations describe unsharp masking as:
   v(m,n) = u(m,n) + λg(m,n), where g(m,n) is a gradient (e.g., discrete Laplacian).

Also, a spatial high-pass filter can be obtained as $h_{HP}(m,n) = \delta(m,n) - h_{LP}(m,n)$. Consider a 3×3 low-pass filter (box filter) with all weights =1/9. (15 Pts)

a. Write the 3×3 mask (convolution kernel) for h_{LP}.
b. Derive the corresponding high-pass mask h_{HP} using $\delta(m,n)$ (center coefficient 1, others 0). Show the resulting 3×3 kernel.
c. Suppose you apply unsharp masking with $\lambda=1$ and g(m,n) = output of this h_{HP} filter. Show that this is equivalent to simply using a single modified convolution kernel. Provide that kernel.

5. You are given two grayscale images below. Perform histogram matching step by step: (20 Pts)

a. Compute the histogram of the input image.
b. Compute the histogram of the reference image.
c. Compute the Cumulative Distribution Function for both images.
d. Map the pixel values of the input image to new values such that its resulting histogram approximates the reference histogram.
e. Show the transformed (matched) image matrix.
f. Plot the following two histograms on the same or separate axes for comparison:

- The histogram of the references image.
- The resulting histogram of the matched image.

Input image:

10	12	10	14	15	13	12	11
14	15	13	12	11	10	14	15
13	12	11	10	14	15	13	12
11	10	14	15	13	12	11	10
15	14	13	12	11	10	15	14
12	11	10	15	14	13	12	11
10	14	15	13	12	11	10	14
13	12	11	10	15	14	13	12

Reference image:

5	6	7	5	6	7	5	6
7	5	6	7	5	6	7	5
6	7	5	6	7	5	6	7
5	6	7	5	6	7	5	6
7	5	6	7	5	6	7	5
6	7	5	6	7	5	6	7
5	6	7	5	6	7	5	6
7	5	6	7	5	6	7	5

Practical (40 Points)

1. Complete "DIP_HW4_Practical.ipynb" notebook